

Full Length Article

KSNet: Advancing autism prediction via KAN-based graph convolution and multi-source data fusion

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ARTICLE INFO

Keywords:

Autism spectrum disorder
Kolmogorov-Arnold networks
Graph convolutional networks
Phenotypic information

ABSTRACT

Integrating functional brain networks and Graph Convolutional Networks (GCN) architecture reveal promising diagnostic capabilities for brain disorders. Prevailing approaches remain constrained when capturing dynamic nonlinearities, characterizing intricate connectomes, and fusing heterogeneous data sources. To address these limitations, we present KSNet, a novel architectural grounded in graph convolutional networks. Our design synergistically combines KAN-derived graph convolutions (KANGCN) with subject-relational convolutions (SRGCN). The KANGCN module employs B-spline functions to enable tractable optimization of high-dimension nonlinear neurography manifolds. We further implement selective node retention to amplify biologically relevant regions and attenuate noisy connections. SRGCN integrates phenotypic features, such as age and gender, with functional connectivity representations derived from KANGCN to construct an inter-subject relationship graph. The model employs a trainable hierarchical architecture that adaptively aggregates multi-scale features, thereby enhancing its discriminative power. Experimental results using the AAL116 and CC200 brain atlases demonstrate notable improvements across key performance metrics. Specifically, KSNet achieved a 13.8% increase in classification accuracy and a 12.4% improvement in the Area Under the Receiver Operating Characteristic Curve (AUC). Remarkably, the biomarkers identified by KSNet align closely with those reported in existing ASD research literature, offering new biologically interpretable support for clinical ASD diagnosis.

1. Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental condition characterized by deficits in social communication, impaired interactions, and repetitive behaviors (Steer et al., 2010). Currently, ASD diagnosis primarily relies on behavioral observation and clinical evaluation. However, this approach lacks objective biological markers and is susceptible to subjective interpretations influenced by clinical experience (Okoye et al., 2023; Qin et al., 2024). The development of functional magnetic resonance imaging (fMRI) has opened new avenues for early ASD detection. fMRI enables non-invasive monitoring of resting-state brain activity and the analysis of functional connectivity between brain regions, thereby revealing network abnormalities in individuals with ASD (Li et al., 2020; Power et al., 2012). Brain network analysis has emerged as a valuable tool for investigating ASD, leveraging connectivity mapping and graph-theoretic metrics to compare network topology between ASD and neurotypical individuals, facilitating the identification of potential biomarkers (Li et al., 2022; Peng et al., 2022). Nevertheless, the heterogeneity of brain networks and the inherent noise

in multi-center datasets pose significant challenges to identifying stable and reliable biomarkers (Wen et al., 2022).

The intersection of bioinformatics and deep learning has emerged as a transformative field, driving significant advancements in the analysis of complex biological data (Yang et al., 2026). Within this broad context, deep learning techniques have been increasingly applied to the diagnosis of ASD based on neuroimaging data in recent years. From a modeling perspective, existing approaches can be broadly categorized into vector-based models, convolutional models on regular grids, and graph-structured models. Each paradigm offers complementary strengths in characterizing brain network properties but also suffers from intrinsic limitations. Multi-layer perceptron (MLP) models provide a simple and efficient baseline for ASD classification by mapping flattened functional connectivity matrices or region-level features to diagnostic labels, enabling the learning of global nonlinear decision functions (Sachdeva et al., 2024). However, the flattening process treats brain regions as an unordered feature set, which weakens explicit modeling of inter-regional topology and disrupts the structural organization of connectivity patterns. Convolutional neural networks (CNNs) exploit spatial

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<https://doi.org/10.1016/j.neunet.2026.108892>

Received 23 August 2025; Received in revised form 19 March 2026; Accepted 22 March 2026

Available online 27 March 2026

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locality on Euclidean grids to extract features effectively, particularly for structural MRI or voxel-wise fMRI data. Nevertheless, applying CNNs directly to connectome data is suboptimal, as enforcing image-like neighborhoods may be inconsistent with the intrinsic non-Euclidean relationships among brain regions (Kaur et al., 2021; Lakhan et al., 2023; Tao et al., 2019). Transformer-based models address long-range dependencies by leveraging self-attention to model global interactions, but their effectiveness depends heavily on the design of connectivity or graph representations and can be sensitive to upstream processing choices (Datta et al., 2022; Deng et al., 2022). Diffusion-based approaches have recently been introduced to alleviate limited-sample issues by iteratively generating or augmenting connectivity representations, but their clinical utility relies on whether synthesized connectomes preserve meaningful patterns without introducing artifacts (Travers et al., 2012). These limitations have motivated the development of topology-aware graph learning frameworks that operate directly on connectome structures for more reliable ASD prediction.

Graph convolutional networks (GCNs) have gained popularity in brain connectome modeling due to their ability to directly aggregate neighborhood information and propagate node features along graph structures (Kipf & Welling, 2016). GCNs preserve brain network topologies while learning discriminative node representations, making them suitable for ASD diagnosis. Prior studies have proposed dynamic graph modeling and multi-view embedding to capture spatiotemporal variations and improve classification. However, existing GCN approaches face several challenges (Guan et al., 2024; Huang & Chung, 2020; Parisot et al., 2017; Wang et al., 2023): reliance on static or shallow graph structures limits the modeling of complex interactions; fixed neighborhood sizes or manually defined thresholds reduce robustness across datasets; and modeling individual variability and group-level relationships remains insufficient, limiting generalization and biologically meaningful. Furthermore, many pipelines collapse BOLD time series into static connectivity before applying message passing dominated by linear transformations with fixed activation functions, which may be inadequate for capturing region-specific nonlinear interactions and can be sensitive to cross-site noise under limited sample sizes. Recent advances in KANs offer a promising alternative by replacing fixed activations with adaptive, spline-based nonlinear transformations. In the realm of biological graph learning, Zhang et al. (2026) demonstrated the efficacy of KAN-enhanced graph attention networks (KA4GANC) in modeling complex gene regulatory networks, showing that KAN-based convolutions can effectively preserve topological structure while learning nonlinear patterns.

To address these critical limitations and inspired by advances in KAN-enhanced biological graph learning, we propose a novel graph convolutional network framework, KSNet, which enhances nonlinear modeling capacity and integrates group-level information in brain network analysis. In the feature extraction layer, KSNet incorporates the KANGCN module, replacing the fixed activation functions in conventional GCNs with learnable B-spline functions based on the Kolmogorov-Arnold representation theorem. This enables each edge to model nonlinear transformations independently, thereby capturing complex functional interactions between brain regions more accurately. Compared to existing KAN-based graph models, KANGCN embeds function approximation within the graph convolution layer, substantially improving nonlinear modeling capacity while preserving the biological plausibility of brain network topology. At the multimodal fusion level, the SRGCN module overcomes the limitations of conventional feature concatenation by encoding phenotypic information (e.g., age and sex) into a subject-to-subject affinity graph, which is then fused with functional connectivity similarity to form a relational population graph. SRGCN employs hierarchical graph convolution and learnable weight aggregation to adaptively integrate representations from multiple levels, enabling deep fusion of functional imaging and phenotypic data. The key contributions of this study can be summarized as follows:

- (1) We propose KANGCN, which incorporates Kolmogorov–Arnold Networks for interpretable function approximation, improving the modeling of nonlinear brain connectivity patterns during feature propagation.
- (2) We introduce a node selection mechanism to sparsify the graph and identify discriminative brain regions, providing quantitative evidence for ASD biomarker discovery.
- (3) SRGCN constructs cross-individual heterogeneous graphs by combining phenotypic attributes and connectivity-derived representations, employing a hierarchical fusion strategy to achieve deep integration of multi-source information, thereby enhancing the model’s ability to capture discriminative ASD patterns.
- (4) The proposed KSNet framework provides strong insights by identifying ASD-relevant brain regions whose localization aligns closely with key biomarkers reported in existing medical studies.

The remainder of this paper is organized as follows. Section 2 provides a brief overview of the scientific background. Section 3 presents a detailed description of the proposed method. In Section 4, we evaluate the effectiveness of the key components of KSNet through ablation studies and compare the model with several existing ASD prediction methods. Finally, Section 5 concludes the paper and outlines directions for future work.

2. Technical background

2.1. Kolmogorov-Arnold networks

KANs (Liu et al., 2024) have recently garnered significant attention in machine learning research. This theorem demonstrates that any continuous multivariate function can be represented as a finite composition of univariate functions and additive operations. Fig. 1 provides an intuitive comparison between the architecture of MLPs and KANs. Unlike conventional MLPs that employ fixed activation functions on weighted input sums, KANs explicitly decompose high-dimensional functions into hierarchically organized univariate operations. This structural innovation enables both theoretical advantages in efficient complex mapping representation and deeper insights through transparent compositional architecture. The theorem states that any continuous multivariate function $f : [0, 1]^n \rightarrow \mathbb{R}$ can be expressed as a sum of univariate functions, as shown in Equation (1):

$$f(x_1, x_2, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \psi_{q,p}(x_p) \right) \quad (1)$$

Here, Φ_q and $\psi_{q,p}$ denote continuous univariate functions. Each $\psi_{q,p}$ nonlinear transformation applies to input variable x_p , while Φ_q nonlinearly combines these transformed inputs to construct sophisticated mappings. The theorem’s significance lies in demonstrating that any multivariate function can be represented through additive compositions of univariate functions, thereby circumventing the need for explicit high-dimensional mappings. In KANs, B-spline functions are used to parametrize the univariate functions Φ_q and $\psi_{q,p}$, as detailed in Equation (2):

$$\phi(x) = \sum_{i=1}^N c_i \cdot B_i^k(x) \quad (2)$$

The formulation employs $B_i^k(x)$, representing k th order B-spline basis functions, where trainable coefficients govern spline shape modulation, and N denotes the number of control nodes. This spline-based parameterization enables interpretable, computationally efficient, and precise representation of univariate functions in KANs, demonstrating particular efficacy in scientific computing applications and physics-informed modeling scenarios.

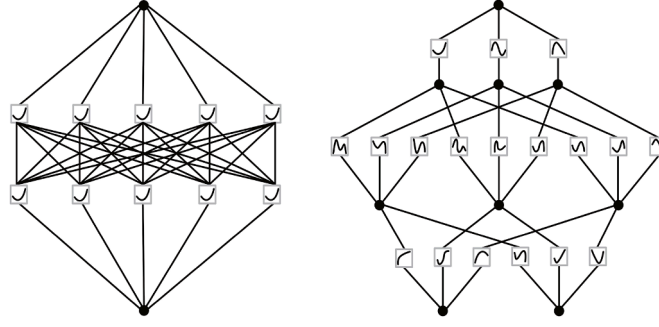


Fig. 1. Comparison of the reasoning process structure of MLPs and KANs.

2.2. Graph convolution network

GCNs (Zhang et al., 2022a) represent a specialized deep learning framework for graph-structured data, employing localized operations to analyze complex node relationships and structural patterns. The core concept extends conventional grid-based convolution to irregular graph structures through neighborhood information aggregation. Central to GCN architecture are graph convolution layers that iteratively update node representations by combining features from a node and its immediate neighbors. For a graph $G = (V, E)$ with node set V and edge set E , let $X \in \mathbb{R}^{N \times D}$ denote the node feature matrix where N is the total nodes and D is the feature dimensions. The adjacent matrix $A \in \mathbb{R}^{N \times N}$ encodes inter-node connectivity. The GCNs formulation can be mathematically expressed as Eq. (3):

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right) \quad (3)$$

Here, $\tilde{A} = A + I_N$ represents the adjacency matrix with self-connections, where I_N denotes the identity matrix. The degree matrix $\tilde{D}$ is diagonal with entries $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. $W^{(l)}$ corresponds to the trainable weight matrix at layer l , and $\sigma(\cdot)$ indicates the non-linear activation function. This formulation enables hierarchical aggregation of multi-hop neighborhood information across network layers. The resulting node embeddings capture discriminative patterns of inter-regional connectivity and functional relationships in brain networks, establishing a powerful framework for learning clinically relevant features in neuropsychiatric disorder diagnosis.

3. Method

3.1. Overview

This study presents a Multi-Source Feature Fusion framework that integrates fMRI and phenotypic data for ASD prediction and the identification of potential neurobiological biomarkers. As illustrated in Fig. 2, The framework systematically implements, including functional network construction from BOLD signals, feature abstraction via KANGCN, relational modeling and embedding learning via SRGCN, and final diagnostic classification. The workflow begins with deriving nodal feature matrices from preprocessed BOLD time series $T \in \mathbb{R}^{n \times l}$, where n denotes ROIs and l temporal observations, to construct whole-brain connectivity networks. The KANGCN module then performs multi-hop feature propagation across network nodes, progressively encoding high-order connectivity patterns through successive graph convolution layers. A saliency-driven node selection mechanism subsequently identifies discriminative brain regions using feature importance metrics, effectively suppressing noise while preserving relevant nodes. Phenotypic attributes are fused with KANGCN-generated global representations to construct a subject-wise relational graph that encodes demographic and neurofunctional correlations. The SRGCN module ultimately derives compact embeddings from the subject relationship network, achieving dual objectives

of ASD diagnosis and biomarker discovery through analysis of clinically salient graph patterns.

3.2. KAN graph convolutional network

3.2.1. Construction of brain networks

Current methods for constructing brain networks predominantly rely on Pearson correlation coefficients between ROIs, which may inadequately capture the brain's complex connectivity patterns (Qu et al., 2021). This linear correlation-based approach overlooks critical nonlinear dynamics and time-dependent features in neural networks, thereby constraining mechanistic understanding and predictive accuracy. To address these limitations, this study employs a framework that integrates a temporal sequence encoder with a FC generator for effective brain network modeling. The temporal encoder employs multi-head self-attention mechanisms to identify long-range dependencies and extract contextually rich features from time-series data (Vaswani et al., 2017). The process operates as follows, for l layer temporal inputs $T \in \mathbb{R}^{n \times l}$, Eq. (1) generates a triplet representation comprising (Q, K, V) .

$$\begin{aligned} Q &= T^l W_q^l \\ K &= T^l W_k^l \\ V &= T^l W_v^l \end{aligned} \quad (4)$$

Where, $W_q^l, W_k^l, W_v^l \in \mathbb{R}^{l \times d}$ represent learned parameters in the linear projection layers, where d notes the channel dimension of the triplet. The single-head attention mechanism is computed as Eq. (5):

$$Att(Q, K, V) = SoftMax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (5)$$

To enhance feature diversity through parallel processing, we extend this mechanism to multi-head attention. The encoder output with multi-head attention is defined by Eq. (6):

$$Encoder(T^{(l-1)}) = Concat(head_1, head_2, \dots, head_h)W_{MLP} \quad (6)$$

Each, $head_j$ is computed as $head_j = Att(T^{(l-1)}W_q^l, T^{(l-1)}W_k^l, T^{(l-1)}W_v^l)$, where h indicates the number of heads. Let denote the output of the temporal encoder, formally expressed a $E = Encoder(X)$.

Using the temporal encoder's output features, the FC generator produces the FC matrix $A = E_A E_A^T$, where $E_A = Softmax(E)$. The resulting matrix $A \in \mathbb{R}^{v \times v}$ is inherently non-negative due to the self-multiplication operation. Simultaneously, the SoftMax function amplifies task-relevant connection weights, ensuring the FC matrix is both task-oriented and compatible with GCNs. Furthermore, the capacity of the generated FC matrix to pinpoint significant abnormal connections within brain networks facilitates the uncovering of potential large-scale functional impairments. Node positional information is critical for modeling complex graph structures. However, existing approaches often overlook this

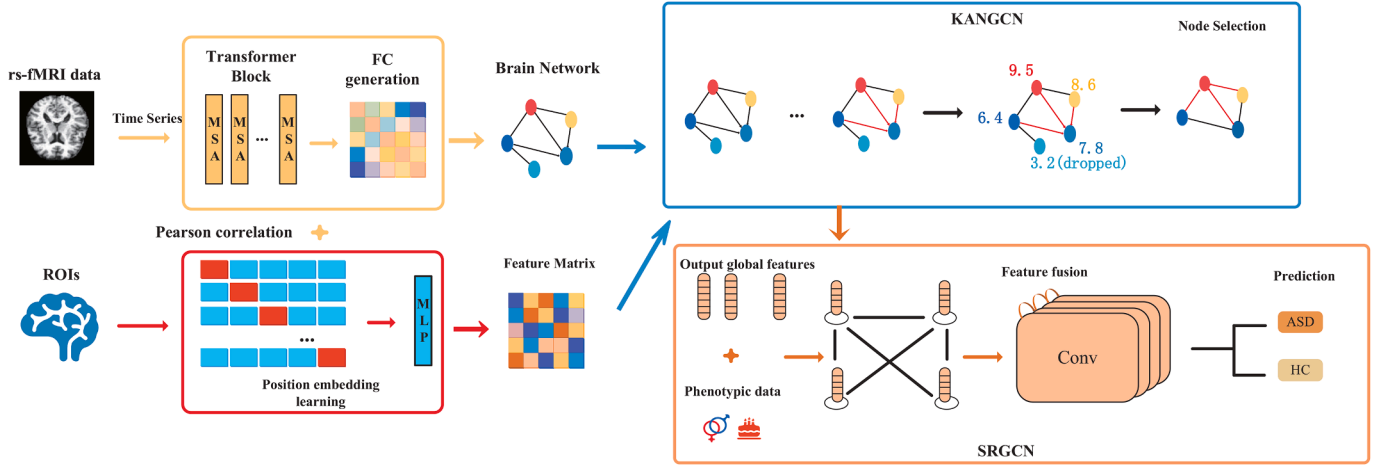


Fig. 2. Overview of KSNet: Time series are input for prediction, and RGTNet selects key brain regions to aid diagnosis.

property in brain network analysis, despite the strong correlation between the anatomical distribution of ROIs and their functional characteristics (Wang et al., 2023). To address this gap, we implement a spatially aware feature enhancement method. First, functional connectivity strength between nodes is computed from BOLD signal time-series using Pearson correlation coefficients, yielding feature vectors $a_i \in \mathbb{R}^d$, where d denotes the feature dimension. To encode spatial information, a one-hot vector is $p_i \in \{0, 1\}^N$ generated for each node v_i , with the i th element set to 1 and others to 0. This vector is linearly transformed into a positional embedding via $p'_i = M_{MLP}(p_i)$. The node's feature vector a_i is then augmented by summing it with its positional embedding $\hat{a}_i = a_i + p'_i$. The final feature matrix $Z \in \mathbb{R}^{N \times d}$ is constructed by aggregating all enhanced node feature vectors.

3.2.2. KANGCN

KANs demonstrate superior ability over MLPs in representing high-dimensional complex functions, thereby providing greater utility for scientific discovery and clinical diagnostics (Bresson et al., 2024; Zhu et al., 2025). By leveraging the Kolmogorov-Arnold representation theorem, KANs decompose nonlinear interactions among high-dimensional brain network nodes into combinations of univariate functions, reducing parameter redundancy and alleviating training challenges. The decomposition framework of KANs provides mathematical clarity through linear combinations of univariate functions, allowing for intuitive visualization of node relationships and connectivity patterns. This approach preserves the biological plausibility of brain network topology while offering a clear and interpretable model of complex brain interactions. While MLPs offer greater flexibility, their reliance on extensive training data and the black-box nature of their architecture often hinder the extraction of clinically actionable insights, making KANs particularly promising for brain network-based disease diagnosis. We introduce an enhanced GCN variant, KANGCN, by replacing standard GCN parameters and ReLU activation with a dedicated KAN layer.

$$H^{(l)} = \Phi^{(l)} \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l-1)} \right) \quad (7)$$

In KANGCN, a third-order B-spline function is used as the basis function to parameterize the univariate functions Φ_q and $\psi_{q,p}$. The grid size is set to 5, corresponding to 8 control nodes evenly distributed in the interval $[-1, 1]$. During forward computation, the aggregated node features are used to calculate the B-spline basis function values $B(x)$ via the Cox-de Boor recurrence formula. These values are then linearly combined with the learnable spline weights W_{spline} , and added to the base linear transformation activated by the SiLU function, i.e.,

$$\Phi(x) = \text{SiLU}(x) \cdot W_{\text{base}} + B(x) \cdot W_{\text{spline}}$$

The base weights W_{base} are initialized using Kaiming uniform initialization, while the spline weights are initialized with small random perturbations. The B-spline function allows for smooth approximation of complex nonlinear relationships while maintaining computational efficiency. This design provides each input-output feature pair with an independent learnable nonlinear transformation capability, enabling interpretable representations of high-dimensional brain network data, and ensuring biological plausibility during the feature propagation process.

KANGCN employs a learnable dynamic node filtering strategy to adaptively compress graph structures by evaluating feature importance. This process enhances model efficiency and performance by focusing on critical nodes. Given a graph with N nodes and their feature matrix $Z = [z_1, z_2, \dots, z_N]^T \in \mathbb{R}^{N \times d}$, where $z_i \in \mathbb{R}^N$ corresponds to the feature vector of the i th node, the node importance scores are computed as follows:

$$s_i = z_i \cdot w + b \quad (i = 1, 2, \dots, N) \quad (8)$$

Where $w \in \mathbb{R}^d$ is a learnable weight vector, $b \in \mathbb{R}$ is a bias term, $\cdot$ denotes the dot product. This operation maps high-dimensional features to a 1D importance space, enabling gradient-based optimization of discriminative feature weights. To mitigate scale variations across graph structures, raw scores are normalized using a Sigmoid function:

$$\hat{s}_i = \frac{1}{1 + e^{-s_i}} \quad (9)$$

$$X' = X[\mathcal{V}', :] \in \mathbb{R}^{K \times d} \quad (10)$$

Normalized scores $\hat{s}_i \in (0, 1)$ quantify node importance. Nodes are ranked in descending order of $\hat{s}_i$, yielding an indexed sequence $\{i_{(1)}, i_{(2)}, \dots, i_{(N)}\}$, where $\hat{s}_{i_{(1)}} \geq \hat{s}_{i_{(2)}} \geq \dots \geq \hat{s}_{i_{(N)}}$. With a retention ratio $\gamma = 0.7$, the top $K = \gamma N$ nodes form the subset $\mathcal{V}' = \{i_{(1)}, i_{(2)}, \dots, i_{(K)}\}$. The feature matrix is then subsampled using these indices, and the adjacency matrix is dynamically compressed to $A' = A[\mathcal{V}', \mathcal{V}']$, completing the graph sparsification. Node importance scores are derived from the feature vectors output by the KAN layer, rather than the original input features. This enables the node selection mechanism to leverage high-level semantic representations learned by KAN, enhancing the model's ability to identify ASD-related brain regions. By integrating KAN's adaptive spline-based representations with dynamic graph sparsification, a structural innovation is introduced, enhancing both model efficiency and capacity for mechanistic discovery. This approach employs a dynamic node selection mechanism that adaptively compresses the graph by evaluating node importance, focusing on critical nodes and discarding irrelevant ones, thereby optimizing performance and promoting scientific clarity. KAN's value in revealing underlying dynamics is

demonstrated in two key aspects: first, spline functions directly capture nonlinear response patterns between brain regions, and second, the node selection process maps the spatial distribution of key regions, thereby validating the model's effectiveness in extracting clinically relevant brain network features.

3.3. Subject relationship graph convolutional network

3.3.1. Construction of subject relationship graph

fMRI provides rich anatomical and functional information, but it exhibits limitations in characterizing inter-subject heterogeneity. Specifically, significant population-level variations in imaging features emerge across age and gender groups (Chaddad et al., 2017). For instance, age-related structural degeneration and gender-specific neural functional distributions may confound disease diagnosis. Furthermore, heterogeneity in physiological traits and pathological progression substantially impacts clinical manifestations and disease trajectories. To address these challenges, we develop a multimodal subject relation network that integrates discrete similarity of phenotypic features with continuous similarity of embedding features (Zhang et al., 2022b), generating a composite adjacency matrix A . The relational matrix is computed through phenotypic feature comparisons. For continuous variables, nodes i and j receive a relationship score increment of 1 if their feature difference falls below a predefined threshold. For categorical variables, a score of 1 is assigned when values match. Simultaneously, we calculate the correlation distance between embedding features z_i and z_j derived from KAGCN training. These embeddings are then transformed into similarity measures using the Gaussian kernel function defined in Eq. (12).

$$d(i, j) = 1 - \text{correlation}(z_i, z_j) \quad (11)$$

$$S_{ij} = \exp\left(-\frac{d(i, j)^2}{2\sigma^2}\right) \quad (12)$$

Here, σ denotes the mean distance. The final graph network is constructed by fusing the two modalities through a product of their similarity scores, followed by filtering edges with high similarity. This integrated network simultaneously preserves explicit associations from phenotypic features and implicit similarities from deep embeddings, thereby providing a unified graph representation for downstream analyses.

3.3.2. SRGCN

An affinity matrix $A \in \mathbb{R}^{N \times N}$ is constructed by SRGCN based on phenotypic features. For each subject pair (i, j) , the affinity score A_{ij} is computed as the sum of four binary similarity indicators: gender (1 if matched), age (1 if $|\Delta \text{age}| < 2$ years), diagnosis (1 if matched), and IQ (1 if $|\Delta \text{IQ}| < 10$). The final adjacency matrix is formed as the Hadamard (element-wise) product of the affinity matrix and the functional-connectivity similarity matrix. Shallow GNNs exhibit limited information aggregation ranges, constraining their ability to model complex dependencies between distant nodes. As network depth increases, repeated message passing homogenizes node features, degrading model expressiveness and classification performance due to the over-smoothing effect (Liu et al., 2020). To address these limitations, we propose SRGCN, a multi-scale architecture that hierarchically integrates neighborhood information. By stacking graph convolution layers, SRGCN progressively expands information aggregation scope while combining local and high-order neighborhood features. To mitigate over-smoothing in deep networks, SRGCN incorporates a learnable fusion mechanism, as shown in Eq. (14), which adaptively weights feature representations from different layers. In addition, batch normalization and dropout are further incorporated to stabilize training, prevent gradient anomalies, and enhance generalization capability. Each graph convolution layer follows a standardized processing sequence defined in Eq. (13). The core convolution operation is sequentially followed by batch normalization, ReLU activation, and dropout regularization. This

structured approach enhances nonlinear modeling capacity while maintaining training stability.

$$H^{(k+1)} = \text{Dropout}(\text{ReLU}(\text{BatchNorm}(H^{(k)}))) \quad (13)$$

The model stores feature outputs from each layer in a dedicated list *layerout*. These hierarchical representations are then adaptively fused through a learnable weight vector $w \in \mathbb{R}^L$ defined in Eq. (14).

$$\alpha_i = \frac{\exp(w_i)}{\sum_{j=1}^L \exp(w_j)}, \quad E = \sum_{i=1}^L \alpha_i \cdot H^{(i)} \quad (14)$$

Here, α_i denotes the adaptive weight assigned to the i th layer, with $H^{(i)}$ representing the corresponding hierarchical feature representation. The fused feature matrix E is subsequently projected into the final output space through a *KAN* layer. This projection yields the classification results via $Y = \text{KAN}(E)$, where Y contains the predicted class probabilities for target categories.

3.4. Loss function

The primary prediction loss employs the cross-entropy loss function:

$$\mathcal{L}_{CE} = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})] \quad (15)$$

Where m denotes the total number of samples, with $y^{(i)}$ representing the true label and $\hat{y}^{(i)}$ the predicted value of the i th sample.

3.5. Group loss

Individuals with ASD frequently exhibit shared functional brain network patterns within the group, though individual variability introduces notable heterogeneity in these manifestations (Hahamy et al., 2015). Based on this prior knowledge, we introduce a group loss function composed of intra-group and inter-group components. The intra-group loss enhances feature similarity among samples from the same diagnostic category while suppressing individual variability effects, thereby improving inter-class discriminability. Given the substantial individual differences within ASD populations, the intra-group loss reduces within-group dispersion in connectivity matrices, effectively mitigating noise contamination and extreme sample biases. Given a class $c \in C$ and the set $S^c = \{i | Y_{i,c} = 1\}$ which involves all the indexed specimens with label c , the means μ_c the means σ_c^2 , of their FC matrices A within a batch, the intra-group loss is formulated as:

$$L_{\text{intra}} = \sum_{c \in C} \sum_{i \in S^c} \frac{\|A^i - \mu_c\|_2^2}{|S^c|} = \sum_{c \in C} \sigma_c^2 \quad (16)$$

The inter-group loss is designed to maximize the distance between different groups, thereby ensuring clear separation among them. This approach enhances the model's ability to distinguish between diagnostic categories by increasing the feature distance across classes. Additionally, it helps mitigate the risk of overfitting by reinforcing inter-group differences, particularly in scenarios with imbalanced samples or high levels of noise. Specifically, given groups a and b , the inter-group loss is calculated by broadening the means μ_a and μ_b as:

$$L_{\text{inter}} = - \sum_{a, b \in C} \|\mu_a - \mu_b\|_2^2 \quad (17)$$

The final group loss is defined as the sum of intra-group and inter-group losses $L_{\text{group}} = L_{\text{intra}} + L_{\text{inter}}$. By reinforcing intra-group consistency and inter-group distinctiveness, the model is better able to extract key features from complex brain network data, thereby improving the accuracy and robustness of autism diagnosis.

3.6. Mutual information loss

KAGCN's node retention capability motivates our adoption of mutual information (MI) as the loss function. This approach maximizes MI between critical nodes and global embeddings while suppressing contributions from non-informative regions, thereby amplifying the weighting of diagnostically relevant brain areas and ensuring task-aligned weight distributions. During classification, the model assigns weights approaching 1 to critical ROIs and near 0 to non-critical regions, creating sharp distinctions that improve diagnostic precision and feature specificity. The formal implementation is detailed in Eq. (18):

$$\mathcal{L}_{MI} = \frac{1}{N} \sum_{i=1}^N n([x_i; y_i]) - \log \left(\frac{1}{M} \sum_{j=1}^M \exp \left(n([x_i; y_j^-]) \right) \right) \quad (18)$$

Here, N denotes the total number of positive example pairs in the current batch, while M represents the number of negative examples associated with each positive pair. The term x_i corresponds to the representative features generated by the model for the node i , whereas y_i reflects its inherent characteristics or broader contextual features. Features from non-relevant examples y_j^- are combined and standardized through a normalization process to quantify their comparative influence.

3.7. Contrast loss

This study introduces a contrastive loss function to optimize representation learning in subject-relation networks, offering two principal advantages. First, it enables comprehensive fusion of phenotypic data and KAGCN-derived global features through a unified joint embedding space. Second, the contrastive learning framework significantly enhances model sensitivity to inter-subject heterogeneity through strategic positive/negative sample pairing. Edge weights guide sample selection: node pairs with strong connections (high edge weights) are designated as positive samples. We minimize Gaussian-based distance metrics between their embeddings to drive similarity scores toward 1. Conversely, weakly connected pairs serve as negative samples, with maximized distances forcing similarity toward 0. The contrastive loss formulation is detailed in Eq. (19):

$$\mathcal{L}_{contrast} = -\frac{1}{|\mathcal{P}|} \sum_{(v_i, v_j) \in \mathcal{P}} \log s(v_i, v_j)^+ + \frac{1}{|\mathcal{N}|} \sum_{(v_i, v_k) \in \mathcal{N}} \log(1 - s(v_i, v_k)^-) \quad (19)$$

Let $s(v_i, v_j)^+$ and $s(v_i, v_k)^-$ denote Gaussian similarity scores for positive and negative sample pairs, respectively, where $\mathcal{P}, \mathcal{N}$ represent the counts of positive and negative samples. The composite loss function is defined as (20):

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{group} + \beta \mathcal{L}_{MI} + \gamma \mathcal{L}_{contrast} \quad (20)$$

Hyperparameters α, β , and γ balance the contributions of individual loss components. Through systematic ablation studies, $\alpha = 0.3$, $\beta = 0.01$, and $\gamma = 0.001$ were empirically selected to optimize performance while preserving interpretable relationships among loss terms. This configuration clarifies the distinct contributions of each component to model behavior.

4. Experiments

4.1. Data acquisition and pre-processing

The KSNNet model was rigorously evaluated on the Autism Brain Imaging Data Exchange (ABIDE) dataset (Di et al., 2014), a multi-institutional initiative encompassing 17 international sites. ABIDE provides open-access resting-state functional MRI (fMRI), anatomical scans, and phenotypic data, offering comprehensive neuroimaging resources

for autism spectrum disorder research. Standard preprocessing was implemented through the Connectome Computation System (CCS). Specifically, we utilized the Preprocessed Connectomes Project (PCP) data generated via the Configurable Pipeline for Connectome Analysis (CPAC), which included slice timing correction, motion realignment, and intensity normalization. This process yielded 1112 high-quality fMRI datasets with phenotypic information, including 539 autism patients and 573 neurotypical controls.

4.2. Experimental setup

The time-series encoder is implemented as a single-layer Transformer with four attention heads, an embedding dimension of 8, Q/K/V projection dimensions of 32, and a dropout rate of 0.1. The KAGCN module comprises three KAN-based graph convolution layers, configured with $grid_size = 5$, $spline_order = 3$, and SiLU as the default activation.

The SRGCN module is implemented as a three-layer GCN with a hidden dimension of 20 and a dropout rate of 0.3; layer-wise outputs are fused through a softmax-weighted, learnable aggregation. The EDGE module is configured with a hidden dimension of 128, and edge weights are computed via cosine similarity. Model optimization is performed with Adam (learning rate 1×10^{-3} , weight decay 1×10^{-4}) with a batch size of 16 for 500 epochs. The overall objective combines the group loss ($\alpha = 0.3$), mutual-information loss ($\beta = 0.01$), and contrastive loss ($\gamma = 0.001$) as weighted terms. Performance is evaluated using five-fold cross-validation, and all experiments are implemented in PyTorch and executed on an NVIDIA RTX 3090 GPU. To assess model generalizability, we employed two distinct brain parcellation schemes. (1) Automated Anatomical Labeling (AAL) atlas with 116 regions. (2) CC200 atlas defining 200 functionally coherent regions via spatially constrained spectral clustering. Performance evaluation employed 5-fold cross-validation with four metrics: Specificity (SPEC), Sensitivity (SEN), Accuracy (ACC), and Area Under the Curve (AUC). All compared algorithms used publicly available implementations under identical software configurations to ensure fair comparison.

$$Sensitivity(Sen) = \frac{TP}{TP + FN} \quad (21)$$

$$Specificity(Spec) = \frac{TN}{FP + TN} \quad (22)$$

$$Accuracy(Acc) = \frac{TN + TP}{FP + TP + FN + TN} \quad (23)$$

Where TP, TN, FN and FP are true cases, true negative cases, false negative cases and false positive cases respectively.

4.3. Ablation experiment

Our KSNNet is mainly composed of KAGCN and global SRGCN components. To verify the effectiveness of the two components, we conducted ablation studies. All experiments were performed under identical hyperparameter settings and five-fold cross-validation.

4.3.1. Ablation study for KSNNet

To systematically evaluate the contribution of each component within KSNNet to the overall classification performance, we performed a series of ablation experiments. In one variant, the KAN-based GCN module was substituted with a standard GCN architecture, allowing us to assess the influence of KAN on feature extraction and representation learning. Another variant excluded the node scoring and filtering mechanism, enabling the evaluation of its role in retaining salient node information and mitigating model complexity. Additionally, the subject relation graph and its corresponding training strategies were removed in a separate model configuration, facilitating an analysis of the phenotypic relationship modeling introduced by the SRGCN module. These

Table 1

Analysis of the impact of different model components on classification performance.

Model	SEN(%)	SPEC(%)	ACC(%)	AUC(%)
w/o KANGCN	84.3	85.7	85.1	91.4
w/o NF	86.1	84.6	85.1	90.7
w/o SRGCN	72.1	72.7	72.0	72.0
KSNet	91.6	88.5	89.7	93.5

ablated configurations provide a basis for understanding the relative importance of each component in the proposed framework. The model variants are defined as follows:

- w/o KANGCN Replaces the KAN-enhanced GCN with a standard GCN.
- w/o NF Removes the node scoring and filtering mechanism.
- w/o SRGCN Excludes the subject relation graph and its optimization.
- KSNet The complete model integrating KANGCN, NF, and SRGCN components.

As demonstrated in Table 1, KSNet achieves state-of-the-art performance across all metrics, SEN and SPE attaining 91.61% and 88.59%. In contrast, the w/o KANGCN variant shows substantially lower performance 84.31% and 85.79%, confirming the critical role of the KAN mechanism in enhancing discriminative power. The w/o NF model showed a slight increase in SEN to 86.16%, but SPE and AUC declined to 84.67% and 90.79%, respectively. This indicates that the node selection mechanism contributes to improving model specificity and overall performance. The w/o SRGCN variant demonstrates the most severe performance degradation across all metrics, underscoring SRGCN's pivotal role in leveraging inter-subject relationships. These findings further validate the indispensability of phenotypic information for autism spectrum disorder classification tasks.

4.3.2. Ablation study for KANGCN

To evaluate the sensitivity of the proposed KANGCN model to its core hyperparameters, we performed a series of controlled experiments varying the number of GCN layers, hidden dimensions, grid size, and spline order. The corresponding classification ACC and AUC are summarized in Table 2.

Effect of GCN Layers. Increasing the number of GCN layers from 2 to 3 led to consistent improvements in predictive performance. Specifically, a 3-layer configuration achieved a maximum of 88.6% ACC and 93.5% AUC. However, deeper architecture also exhibited higher sensitivity to other parameters such as grid size and were more prone to instability.

Effect of Hidden Dimensions. The hidden dimension size plays a crucial role in the model's capacity. Expanding the dimension from 32 to 64 improved both ACC and AUC, while further increasing to 128 yielded the highest performance, indicating that richer node representations enhance learning. Nonetheless, excessive dimensionality may increase the risk of overfitting.

Effect of Grid Size. The grid size determines the resolution of the B-spline kernel in the graph convolution operator. Our results show that a moderate grid size leads to optimal model performance. Larger values, particularly when combined with deeper networks, frequently resulted in numerical instability-NaN values during training, suggesting a tendency toward gradient explosion.

Effect of Spline Order. The spline order governs the smoothness of the filter functions. A higher spline order 3 produced better performance overall, whereas lower orders either degraded the model accuracy or introduced instability under certain configurations.

Taken together, the best-performing configuration consisted of 3 GCN layers, 128 hidden units, a grid size of 5, and a spline order of

3, yielding an accuracy of 88.6% and an AUC of 93.5% without numerical issues. These findings highlight the importance of appropriately balancing architectural complexity and kernel resolution when designing robust graph convolutional networks for brain connectivity analysis.

4.3.3. Ablation study for SRGCN

To investigate the impact of various phenotypic features on the classification performance of the SRGCN module, several comparative experiments were conducted. As demonstrated in Table 3, the baseline configuration using only age and gender achieved the highest accuracy of 89.7% and AUC of 93.5%, whereas the inclusion of IQ or site information led to a significant decline in performance. The reasons for this decline are as follows: With respect to IQ, approximately 30% of the FIQ data in the ABIDE dataset is missing, and the imputation strategy may introduce noise. Moreover, high-functioning autism patients have IQ scores similar to those of healthy controls, which diminishes the discriminative power of this feature. Regarding site information, the sample distribution across the 17 collection sites is highly imbalanced, and the affinity graph, constructed based on site similarity, suffers from sparsity. Furthermore, site information is essentially metadata related to data collection rather than biological features, potentially introducing confounding factors unrelated to diagnosis. Additionally, an excess of phenotypic features can result in a more uniform distribution of affinity scores, reducing the ability to distinguish key features. Therefore, this study utilizes age and gender as the core features for phenotypic data fusion, achieving optimal classification performance while preserving model simplicity.

To analyze the impact of graph construction and model design in the context of SRGCN, we evaluated six representative GNN architectures under three similarity-based edge-weighting schemes: Gaussian, median, and cosine. This ablation framework allows us to disentangle the effects of network topology and architectural choices on overall classification accuracy.

$$\text{Gaussian similarity} = \exp\left(-\frac{\|x_i - x_j\|_2^2}{2\sigma^2}\right) \quad (24)$$

$$\text{Median similarity} = \exp\left(-\frac{\|x_i - x_j\|_2^2}{\sigma(x_i)\sigma(x_j)}\right) \quad (25)$$

$$\text{Cosine Similarity} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \cdot \sqrt{\sum_{i=1}^n y_i^2}} \quad (26)$$

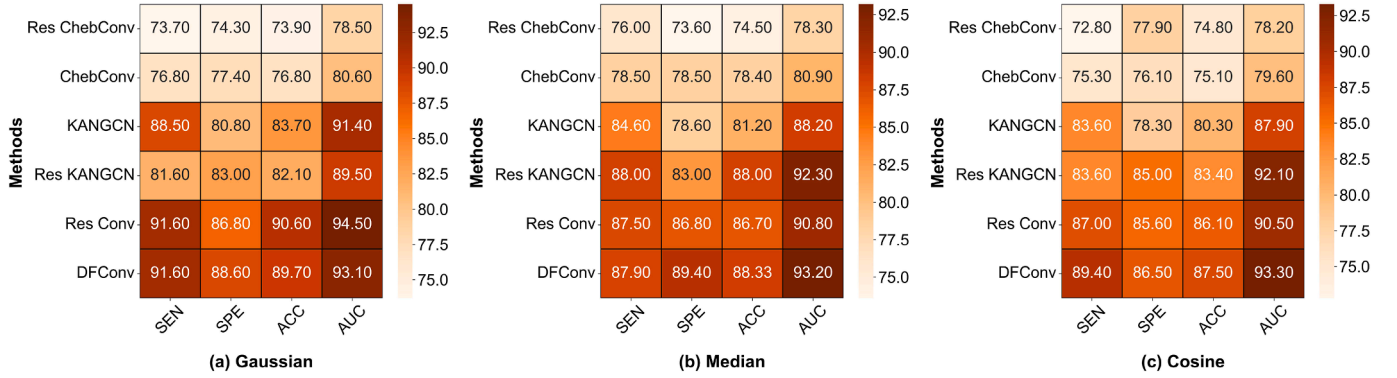
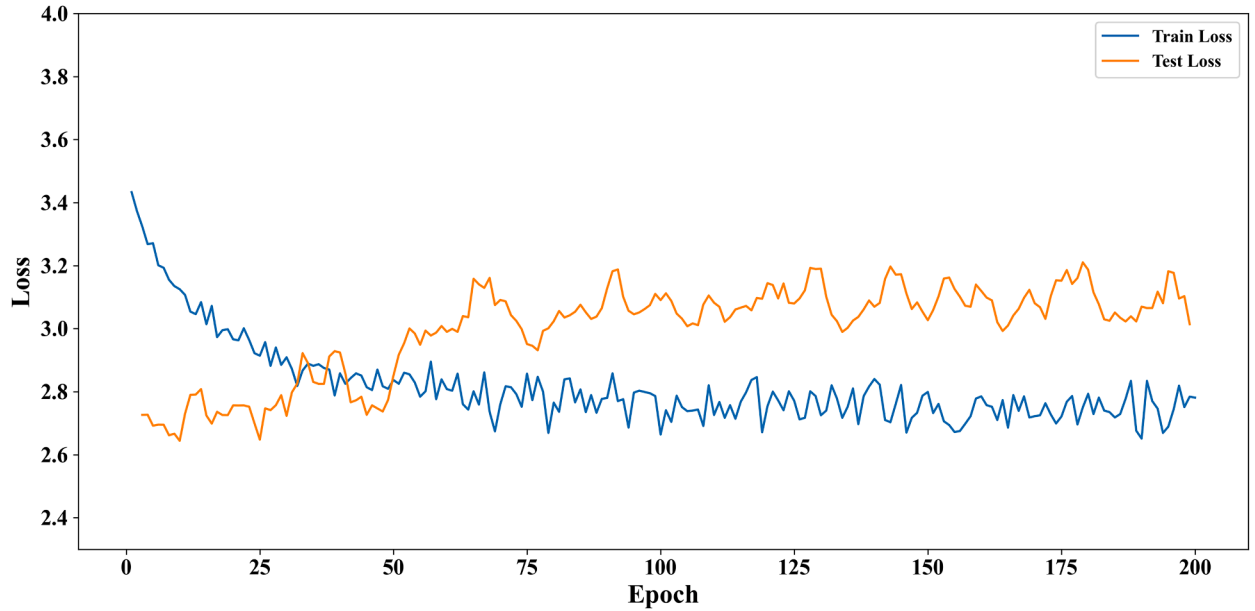
As shown in Fig. 3, the DFConv consistently demonstrated superior performance across all similarity functions. Although its AUC with the Gaussian similarity function was marginally lower than that of ResConv, it achieved the highest AUC of 93.2% when combined with the median similarity function. Under the cosine similarity scheme, it also yielded the best accuracy, reaching 87.5%, further supporting its effectiveness. In addition to overall classification metrics, this architecture performed competitively in both sensitivity and specificity, consistently matching or exceeding those of other models across all weighting strategies. These results highlight the robustness and generalizability of the DFConv-based approach for subject relation network classification.

In summary, the DFConv model demonstrated a clear performance advantage in this study, particularly in terms of accuracy and AUC across different similarity functions. These findings suggest that DFConv offers superior classification capability and generalizability when applied to subject relationship networks, providing a valuable reference for future research. Furthermore, as indicated in the results table, the Gaussian similarity function yielded the best overall performance in weight computation. Therefore, all subsequent experiments employed the Gaussian similarity function for edge weight calculation.

Table 2

The influence of key parameter thresholds in KANGCN on the predictive performance of the model.

GCN Layers	Hidden Dim	Grid Size	Spline Order	ACC (%)	AUC (%)	NaN problem
2	32	3	3	86.4	89.8	✓
2	64	3	3	87.5	89.3	
2	64	5	3	87.7	90.6	
2	64	8	3	86.4	89.1	
2	64	10	3	—	—	
3	64	5	2	87.3	89.5	✓
3	128	5	2	88.2	93.2	
3	128	10	2	—	—	
3	128	5	3	89.7	93.5	

**Fig. 3.** Impact of different SRGCN variants and similarity functions on model performance.**Fig. 4.** Training and test loss evolution curves on CC200 atlas.**Table 3**

Comparison using different phenotypic information.

Model	SEN(%)	SPEC(%)	ACC(%)	AUC(%)
IQ	79.7	78.6	75.2	78.2
Site	78.2	82.3	78.8	80.5
Age	88.1	85.7	86.8	85.3
Gender	82.5	83.2	81.9	86.2
Age + Gender + IQ	82.1	84.3	83.4	88.5
Age + Gender + Site	85.3	81.4	82.9	85.4
Age + IQ + Gender + Site	80.9	82.9	81.3	82.5
Age + gender	91.6	88.5	89.7	93.5

4.4. Comparison with other methods

This study evaluates the performance of two representative machine learning (ML) methods, namely Support Vector Machine (SVM) and Random Forest (RF), in comparison with advanced deep learning (DL) models, including conventional neural networks and graph neural networks. The non-GCN group includes ASD-DiagNet (Eslami et al., 2019) and MC-NFE (Wang et al., 2022), while the GCN group comprises BrainGNN (Li et al., 2021), MVS-GCN (Wen et al., 2022), FBNETGEN (Kan et al., 2022), and PLSNet (Wang et al., 2023). As shown in Table 4, KSNet outperformed all competitors with an accuracy of 89.7%.

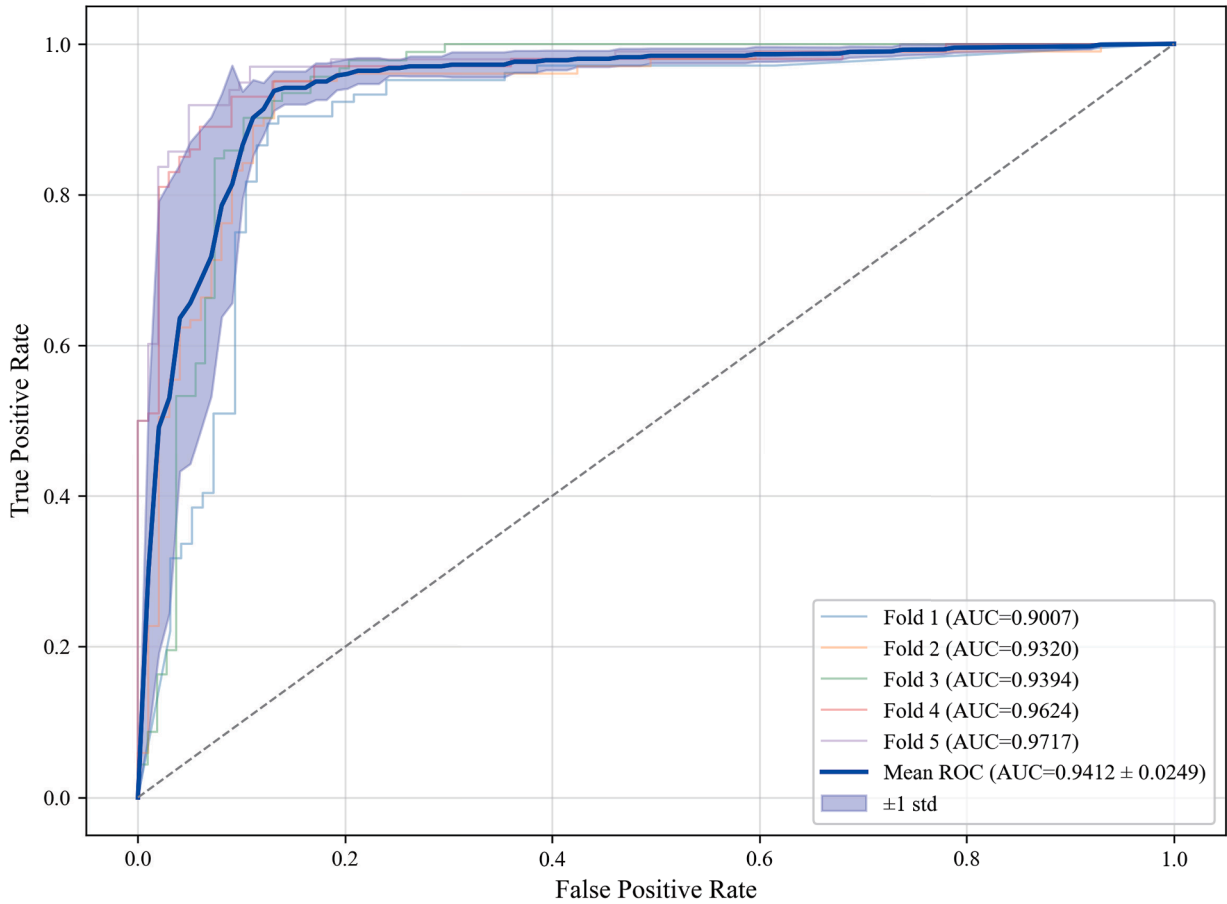


Fig. 5. The ROC curve based on CC200 using 5-fold cross-validation.

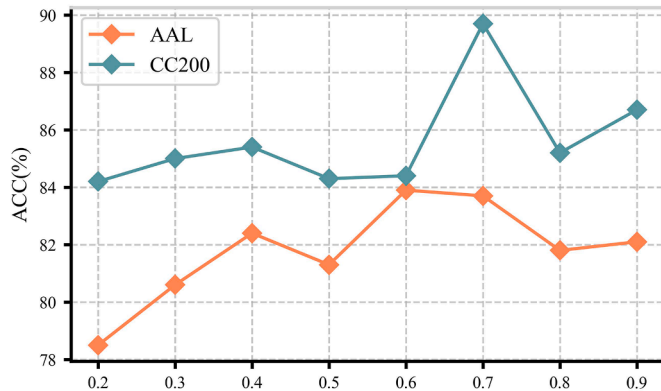


Fig. 6. The impact of different node selection ratios on KSNet performance.

We conduct a detailed comparison of KSNet with various baseline methods. Traditional machine learning methods like SVM and RF flatten the functional connectivity matrix into one-dimensional vectors, losing topological information, with accuracies of only 68.2% and 61.9% on the CC200 atlas. In contrast, ASD-DiagNet and MC-NFE achieve accuracies of 70.3% and 70.2%, respectively, by using autoencoders and multi-channel fusion, but their fully connected architectures fail to model adjacency relationships between brain regions. GCN-based methods, such as BrainGNN, use ROI-aware graph convolution, achieving 61.8% accuracy, but their fixed message-passing mechanisms are ill-suited for the heterogeneous patterns of ASD brain networks. MVS-

GCN improves accuracy to 69.4%, introducing multi-view and graph structure learning, but it only concatenates phenotypic data as auxiliary features, missing structured relationships between subjects. FBNETGEN dynamically generates brain networks, achieving 73.9% accuracy, but is limited to single-subject analysis. PLSNet, with its temporal encoder for BOLD signal dynamics, achieves 75.9% accuracy, but its fixed activation function and lack of phenotypic data integration constrain its performance.

KSNet overcomes the fixed activation function bottleneck with the learnable spline function of KANGCN and achieves structured fusion of phenotypic and imaging data through SRGCN's subject relationship graph. Accuracy on the CC200 atlas improves to 89.7%, a 13.8 percentage point increase over PLSNet. To comprehensively evaluate the model, we analyzed both its training dynamics and its final classification performance. With the increase in training epochs, the variations in the loss for both the training and test sets are depicted in Fig. 4, the evolutionary trajectory of loss on both the training and test sets as training epochs increase. The training loss exhibits a consistent declining trend, indicating that the model effectively fits the feature distribution of the training data. The sharp increase in test loss during the initial stage can be attributed to an overestimation of generalization performance due to the limited scale of the early test samples. Subsequently, after reaching a peak, the test loss enters a fluctuating phase, reflecting that the model maintains a certain degree of generalization capability on unseen samples. Furthermore, the ROC curve in Fig. 5 demonstrates the model's excellent discriminative power, with the Area Under the Curve (AUC) approaching 1.0. The model achieves ideal sensitivity while maintaining a low false positive rate, which fully confirms its precision in identifying positive samples of the target disorder.

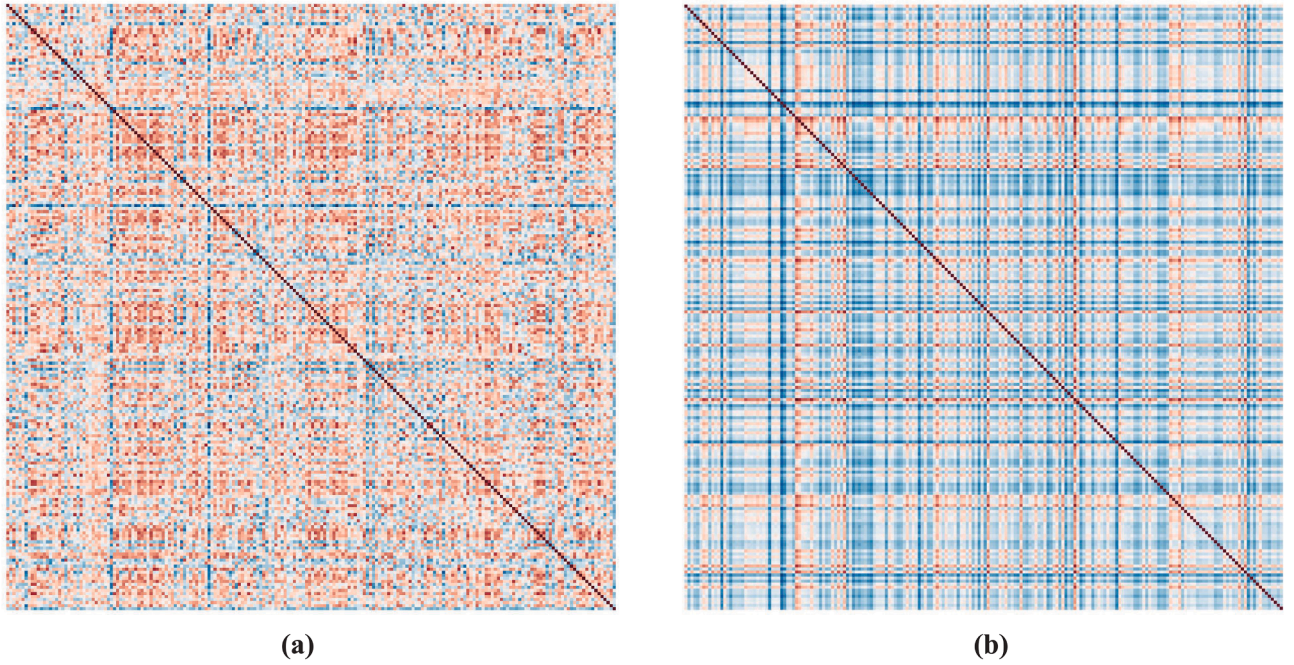


Fig. 7. Comparison of functional connectivity matrices based on Pearson correlation and the representations learned by the proposed KSNet model. (a) Pearson correlation (b) KSNet.

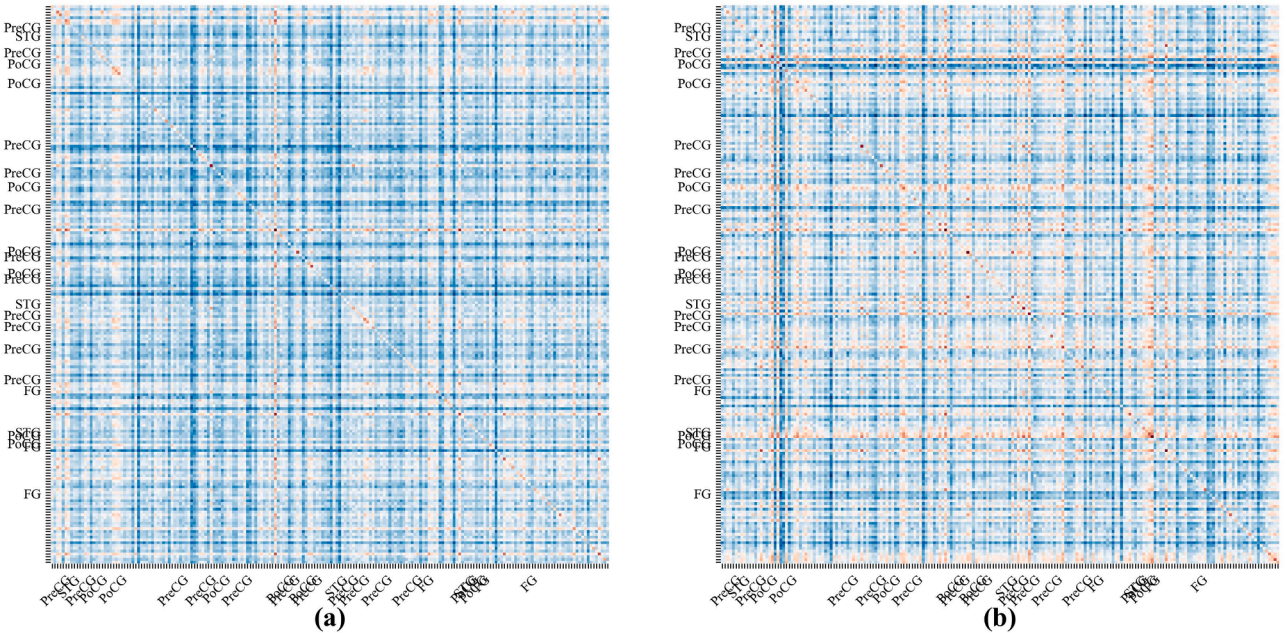


Fig. 8. FC matrices were constructed by the proposed model for (a) NC and (b) ASD subjects.

4.5. Analysis of robust

To validate the stability and consistency of the model in ASD classification tasks, a sensitivity analysis on node selection was conducted, verifying the model's robustness to hyperparameter settings through experiments with varying retention ratios. Fig. 6 depicts the key brain regions selected by the model at different retention ratios (γ) and their stability across various samples. The results show that the model performs optimally and remains stable when retaining 50%–70% of regions, with the selected key regions showing significant overlap with those associated with ASD in existing literature. When the retention ratio exceeded 80%–90%, classification performance declined for both atlases com-

pared to retaining 70% of the nodes. This suggests that low-importance nodes may introduce noise. Therefore, in brain network analysis, retaining more nodes does not necessarily lead to better results, and the optimal selection ratio should be determined based on the specific brain atlas employed.

Beyond the node selection ratio, the choice of brain atlas also plays a critical role. Models utilizing the CC200 functional atlas consistently outperform those based on the AAL anatomical atlas in both traditional and deep learning approaches. This performance gap can be attributed to the fundamental differences in their parcellation strategies. With approximately twice the number of nodes, CC200 provides a more detailed representation of functional connectivity. More importantly, as a



Fig. 9. KSNet identifies biomarkers in the CC200 brain atlas. The figure shows key regions of interest (ROIs) and their color labels.

Table 4

Comparison of predictive performance on the AAL and CC200 brain atlases. Optimal values are highlighted in bold.

ROIs	Model	SEN (%)	SPE (%)	ACC (%)	AUC (%)
AAL116	SVM	56.5	65.9	66.3	67.4
	RF	69.8	58.1	63.4	60.7
	ASD-DiagNet	62.2	64.1	65.9	66.1
	MC-NFE	70.1	63.6	68.4	69.3
	BrainGNN	60.6	62.5	59.0	59.5
	MVS-GCN	68.7	62.4	67.1	66.9
	FBNETGEN	70.6	71.3	70.9	70.4
	PLSNet	72.5	73.2	72.9	74.8
	KSNet	79.8	71.9	83.7	89.1
CC200	SVM	57.5	68.7	68.2	67.4
	RF	46.1	62.0	61.9	60.8
	ASD-DiagNet	68.3	72.2	70.3	72.9
	MC-NFE	70.4	69.2	70.2	71.9
	BrainGNN	61.7	60.8	61.8	60.8
	MVS-GCN	69.8	64.5	69.4	69.0
	FBNETGEN	74.4	71.0	73.9	72.8
	PLSNet	80.0	72.2	75.9	81.1
	KSNet	91.6	88.6	89.7	93.5

functional atlas, it offers higher-resolution parcellations with greater functional homogeneity, which more accurately capture the dynamic functional interactions between brain regions—a key characteristic of neurodevelopmental disorders like ASD. In contrast, AAL's primary focus on anatomical boundaries may fail to capture these nuanced functional dynamics, thus limiting its applicability in ASD research. Therefore, the enhanced functional fidelity of the CC200 atlas facilitates more effective model training and improves predictive performance.

4.6. Analysis of hyper-parameters

This study systematically evaluates three hyperparameters α , β and γ in the loss function to assess their impact on model performance. The group loss α primarily reduces intra-group variations in functional connectivity matrices while enhancing inter-group distinctions, thus receiving the largest weighting range. The mutual information loss β , though less impactful on overall classification, critically improves node importance differentiation, warranting moderate weighting. The Contrast loss γ focuses on fine-tuning subject relationship graphs and maintains minimal weighting to prevent performance interference.

In the first experiment, we analyzed the impact of α when $\beta = 0$ and $\gamma = 0$ fixed. As shown in Table 5, α significantly influenced SEN and ACC. The highest SEN of 90.2% was achieved at $\alpha = 0.3$, with ACC also main-

taining a competitive value of 89.2%. Thus, $\alpha = 0.3$ was selected as the default. For β , with $\alpha = 0.3$ and $\gamma = 0$, increasing led to a slight decline in ACC. At $\beta = 0.05$, ACC dropped to 88.2%, while $\beta = 0.01$ yielded the highest ACC of 89.1%, with SEN and SPE nearing optimal levels. This confirms the beneficial role of mutual information loss, leading to the selection of $\beta = 0.01$ as the default. Finally, γ critically affected the model's generalization and classification performance. At $\gamma = 0.001$, the model achieved a peak of ACC of 89.7% and SEN of 91.6%, indicating enhanced classification capability. Consequently, $\gamma = 0.001$ was chosen to optimize accuracy and stability.

In summary, the optimal hyperparameters $\alpha = 0.3$, $\beta = 0.01$ and $\gamma = 0.001$ balance sensitivity, specificity, and accuracy while improving model stability and generalization.

4.7. Computational complexity analysis

To assess the practical viability of KSNet, model size and computational efficiency were evaluated on the CC200 atlas. KSNet contains approximately 3.62 million parameters, of which 3.52 million are concentrated in the KANGCN predictor. The time-series encoder contributes about 10.7k parameters, while SRGCN account for approximately 24.0k parameters, respectively; the remaining parameters are associated with the fusion and classification components. In terms of computational complexity, a single forward pass requires approximately 0.43 GFLOPs. All experiments were conducted on an NVIDIA RTX 3090 GPU with a batch size of 16 for 500 epochs. Training took about 3 s per epoch on average, corresponding to a total runtime of about 2.8 h. During inference, only the forward pass was timed, and offline procedures were excluded; under this setting, inference required approximately 19 ms per subject. Overall, KSNet achieves fast inference with a moderate parameter budget and computational cost, supporting its suitability for downstream deployment while maintaining strong classification performance.

4.8. Visualization analysis

The FC matrix is pivotal in model training and serves as a key basis for clinical diagnosis. To this end, we visualized the FC matrix learned by the KSNet model. We compared it with the matrix generated using the conventional Pearson correlation method to highlight differences in connectivity patterns. As shown in Fig. 7, panel (a) depicts a brain functional connectivity network constructed using Pearson correlation coefficients without thresholding. Most nodes exhibit strong correlations, which may introduce noise and compromise the accuracy of disease identification. In contrast, panel (b) presents the functional connectivity network produced by the KSNet model, which captures long-range

Table 5
Analysis of hyper-parameters on CC200 atlas (%).

α				β				γ			
α	SEN	SPE	ACC	β	SEN	SPE	ACC	γ	SEN	SPE	ACC
0	88.8	88.4	88.4	0	86.1	90.6	88.2	0	85.2	86.9	87.2
0.1	86.6	90.5	88.3	0.01	90.2	89.2	89.1	0.0001	90.3	86.4	88.1
0.2	89.4	88.4	88.7	0.02	88.3	86.4	87.0	0.0002	88.6	87.8	88.2
0.3	90.2	88.6	89.2	0.03	86.9	88.6	87.8	0.0003	90.6	84.8	88.5
0.4	84.9	91.2	87.9	0.04	90.7	84.5	87.6	0.0004	87.2	84.8	85.7
0.5	86.1	90.6	88.2	0.05	84.8	87.5	86.2	0.0005	87.6	90.7	88.7
1	89.3	90.3	89.4	0.1	87.9	89.3	88.0	0.001	91.6	88.6	89.7

dependencies in the time series to construct the FC matrix. This network exhibits a markedly different structure from that in panel (a). The results indicate that KSNNet effectively identifies discriminative connectivity patterns, enhancing the separability between the two groups and demonstrating the model's advantages in brain network analysis.

To further identify clinically relevant biomarkers for ASD, we visualized and compared the functional connectivity of ASD and HC samples, as shown in Fig. 8. Notably, several functional connections are significantly stronger in the ASD group than in the NC group, revealing distinct differences in connectivity patterns between the two. Increased connectivity is particularly evident in regions such as the precentral gyrus, supramarginal gyrus, lateral occipital cortex, and middle frontal gyrus. Prior research (Nebel et al., 2014) has reported that individuals with ASD exhibit altered connectivity in the precentral gyrus, especially showing increased interactions between the dorsomedial (DM) and ventrolateral (VL) areas. Anderson et al. (2011) found that interhemispheric connectivity is significantly reduced in ASD within functionally relevant regions, including the sensorimotor cortex, superior temporal gyrus, and fusiform gyrus. For improved clarity, we visualized these key brain regions in Fig. 9. The ROIs were highlighted using specific colors based on the CC200 brain atlas. This visualization, combined with consistency across medical findings, confirms that KSNNet produces reliable results and may help identify potential biomarkers to support clinical diagnosis.

5. Conclusion

This paper presents KSNNet, a graph neural network framework for ASD diagnosis and biomarker identification, integrating brain network analysis with phenotypic information through two modules—KANGCN and SRGCN. KSNNet significantly outperforms existing ASD diagnostic methods on the ABIDE dataset, including both traditional machine learning and GCN-based approaches. Ablation analyses indicate that KANGCN primarily enhances intra-graph feature representation through nonlinear modeling, while SRGCN improves robustness and phenotype-driven generalization by incorporating cross-subject phenotypic relationships, together contributing to superior classification performance. The framework demonstrates strong diagnostic accuracy across two brain atlases (AAL116 and CC200), effectively identifies critical brain regions, and reduces noise interference. Although small-sample scenarios were not explicitly evaluated, the use of a multi-center dataset comprising data from 17 institutions inherently reflects inter-site heterogeneity. Future work will explore small-sample and multimodal data, refine graph construction and loss functions, and develop more transparent and clinically actionable models to further improve diagnostic accuracy and clinical applicability.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, we used ChatGPT to improve its readability and language. After using this tool, we reviewed and edited

the content as needed and took full responsibility for the content of the published article.

CRedit authorship contribution statement

Chuang Wang: Writing – original draft, Visualization, Validation, Data curation; **Dongyan Li:** Writing – review & editing, Supervision, Resources, Project administration; **Sixiang Sun:** Resources, Formal analysis; **Junli Liu:** Investigation, Conceptualization; **Yuanyuan Li:** Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is supported by Basic Scientific Research Projects of the Liaoning Provincial Department of Education (JYTMS20230011).

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